SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2- INTREPRETATION

TASK (3 Questions)

Course Code:	DSA0607	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO2 – Precision (BL3)	**Assessment:**	Assessment Tool 2 – Intrepretation
Weightage:	40% of CIA	**Total Marks:**	50 (5 Questions × 10 Marks)
CO2 Statement:	*Apply appropriate visualization methods to represent distributions, proportions, and relationships in data.(BL3)*		
Student Name:	v.surya satvika	**Reg. No.:**	192524254
Section / Batch:	2024/D	**Date:**	

Code of Conduct

I V.Surya Satvika(192524254) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *v.surya satvika*

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Categorical Data Visualization	Comparing category-wise revenue, proportions, and seasonal trends using bar, grouped, and stacked charts	1	10
Distribution Analysis	Understanding defect distribution and variability using histograms, boxplots, and violin plots	2	10
Relationship Visualization	Exploring relationships between variables using scatter plots, heatmaps, and multivariate techniques	3	10
Geospatial & Temporal Visualization	Analyzing delivery performance using maps, time-series plots, and heatmaps	4	10
Distribution Comparison	Comparing patient data distributions using ridgeline plots, density plots, and boxplots	5	10
		GRAND TOTAL:	**50 Marks**

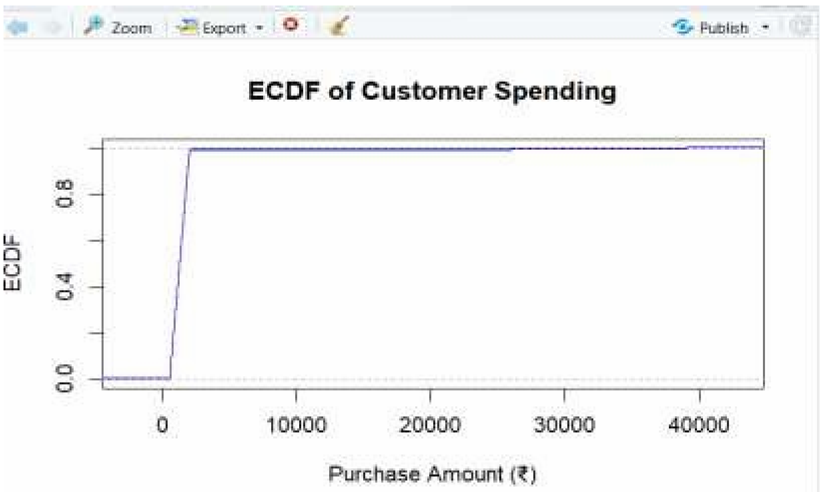
1.An e-commerce company is analyzing the purchase amounts of 10,000 customers during a festive sale. Most customers spent between ₹500 and ₹2,000, while a small number of customers made purchases above ₹25,000. The company wants to understand customer spending behavior, identify high-value customers, and determine whether the spending distribution follows a normal pattern. How would you use ECDF, histograms, and Q-Q plots to analyze the spending data, and what insights could be derived from the results?

Code:

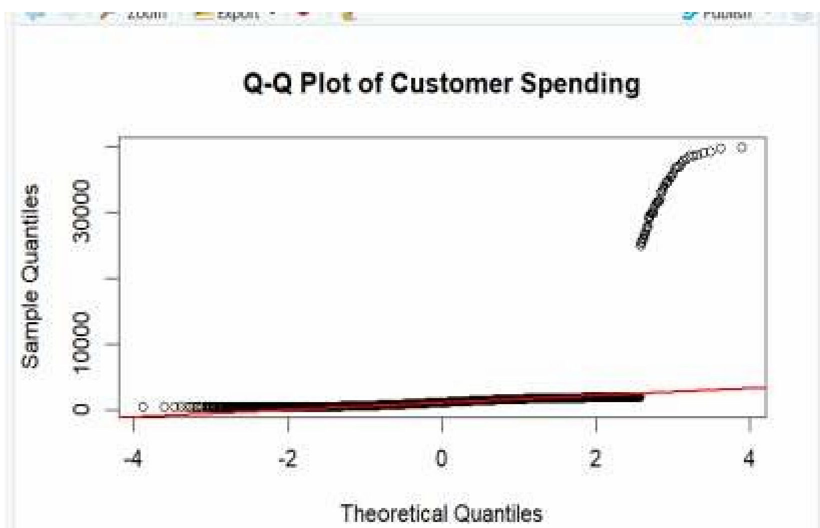
```
# Customer Spending Analysis

set.seed(123)

spending <- c(runif(9950,500,2000),
          runif(50,25000,40000))

# Histogram
hist(spending,
   main="Customer Spending Distribution",
   xlab="Purchase Amount (₹)",
   ylab="Frequency",
   col="skyblue",
   border="black")

# ECDF
plot(ecdf(spending),
   main="ECDF of Customer Spending",
   xlab="Purchase Amount (₹)",
   ylab="ECDF",
   col="blue",
   lwd=2)

# Q-Q Plot
qqnorm(spending,
    main="Q-Q Plot of Customer Spending")
qqline(spending,
    col="red",
    lwd=2)
```

Output:

Customer Spending Distribution
Frequency
0
2000
6000
10000
0
10000
20000
30000
40000
Purchase Amount (₹)



Zoom
Export
Publish
ECDF of Customer Spending
ECDF
0.0
0.4
0.8
0
10000
20000
30000
40000
Purchase Amount (₹)



Q-Q Plot of Customer Spending
Sample Quantiles
0
10000
30000
-4
-2
0
2
4
Theoretical Quantiles

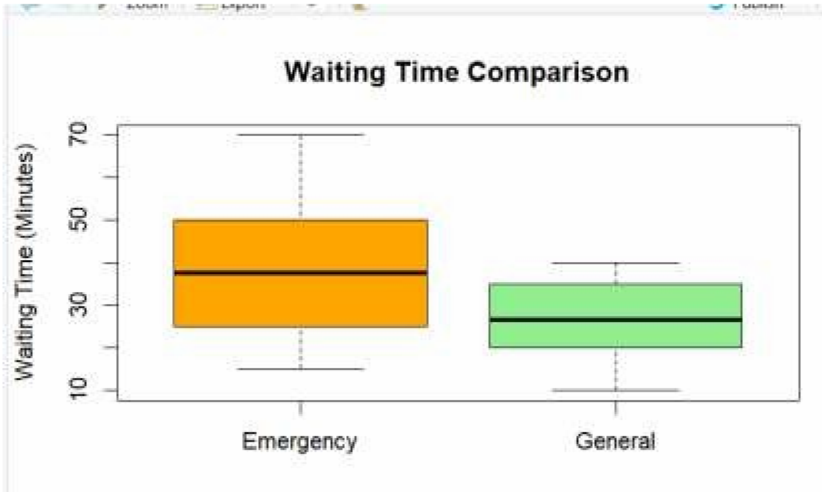
2.A hospital is comparing patient waiting times in two departments: Emergency Care and General Consultation. The management wants to determine whether waiting times in both departments follow similar distributions and whether one department experiences significantly longer delays. Explain how Q-Q plots, boxplots, and density plots can be used to compare the two datasets and evaluate distributional differences.

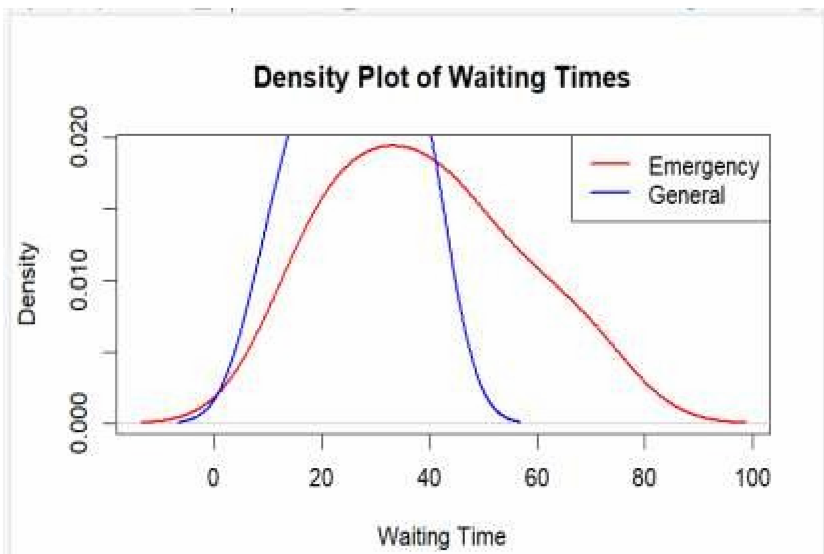
Code:

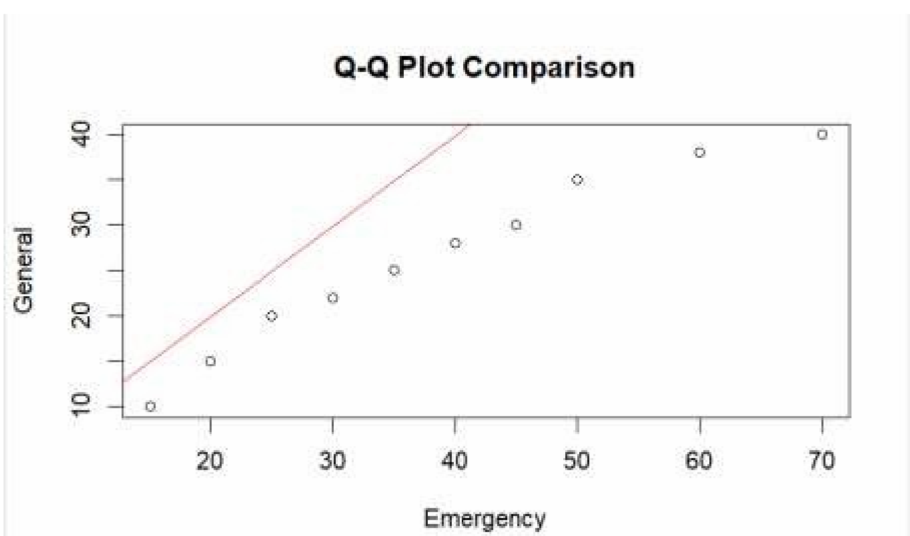
```
# Hospital Waiting Time Analysis

Emergency <- c(15,20,25,30,35,40,45,50,60,70)

General <- c(10,15,20,22,25,28,30,35,38,40)

# Boxplot
boxplot(Emergency, General,
        names=c("Emergency","General"),
        main="Waiting Time Comparison",
        ylab="Waiting Time (Minutes)",
        col=c("orange","lightgreen"))

# Density Plot
plot(density(Emergency),
     main="Density Plot of Waiting Times",
     xlab="Waiting Time",
     col="red",
     lwd=2)

lines(density(General),
      col="blue",
      lwd=2)

legend("topright",
       legend=c("Emergency","General"),
       col=c("red","blue"),
       lwd=2)



# Q-Q Plot
qqplot(Emergency,
       General,
       main="Q-Q Plot Comparison",
```

```
      xlab="Emergency",
      ylab="General")
abline(0,1,col="red")
```

Output:





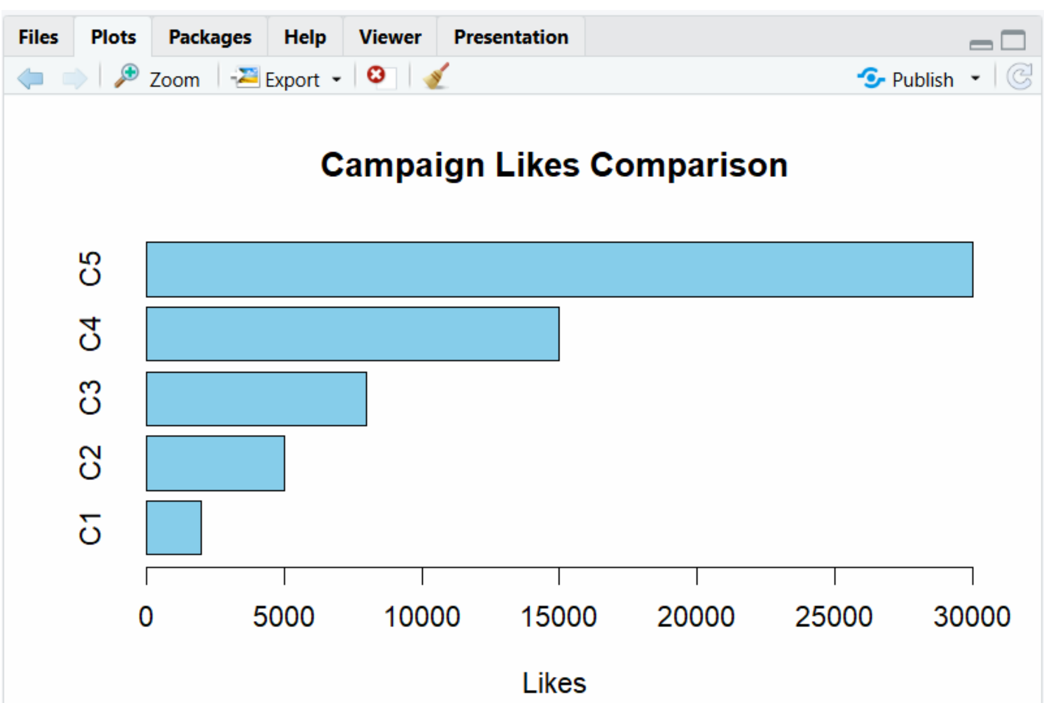


3.A digital marketing team wants to present the number of likes received by five social media campaigns: 2,000, 5,000, 8,000, 15,000, and 30,000 likes. One analyst suggests using a 3D exploded pie chart, while another recommends a horizontal bar chart. Discuss how each visualization may influence audience perception and identify the most effective method for accurately comparing campaign performance. What design principles should be followed to avoid misleading interpretations?

Code:

```
# Campaign Likes

Campaign <- c("C1","C2","C3","C4","C5")

Likes <- c(2000,5000,8000,15000,30000)

# Horizontal Bar Chart

barplot(Likes,
        names.arg=Campaign,
        horiz=TRUE,
        col="skyblue",
        main="Campaign Likes Comparison",
        xlab="Likes",
        border="black")
```

output:



Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context		
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts		
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution		
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools		
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis		

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________